W3C Agents on The Web (WebAgents) CG

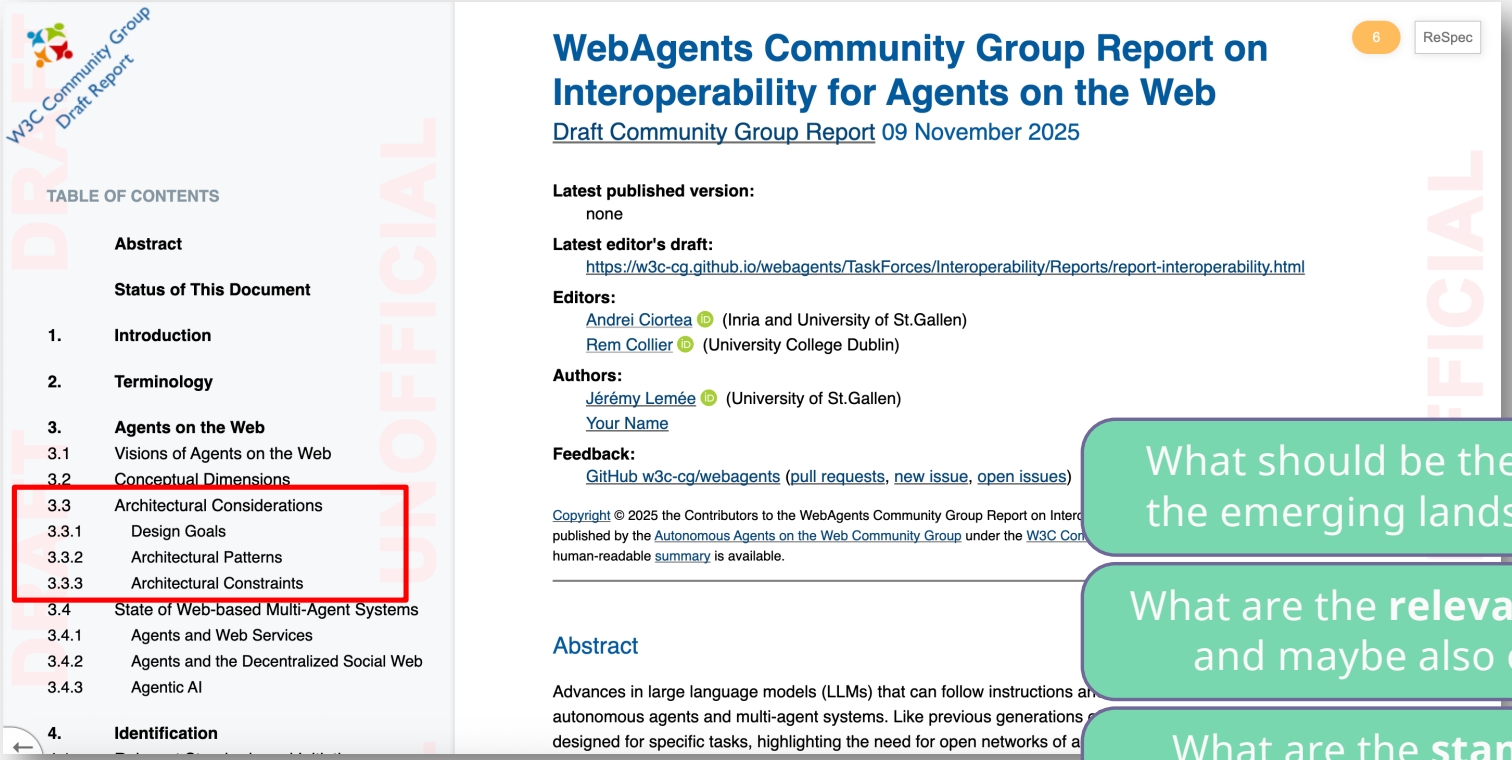
The Web as a Platform for Agents

Architectural design ideas from the Interoperability Report

Andrei Ciortea, **Rem Collier**, Ege Korkan, and Antoine Zimmermann



Interoperability Report



W3C Community Group Draft Report

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WebAgents Community Group Report on Interoperability for Agents on the Web

Draft Community Group Report 09 November 2025

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Feedback:
GitHub w3c-cg/webagents (pull requests, new issue, open issues)

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human-readable summary is available.

Abstract

Advances in large language models (LLMs) that can follow instructions a
autonomous agents and multi-agent systems. Like previous generations
designed for specific tasks, highlighting the need for open networks of a

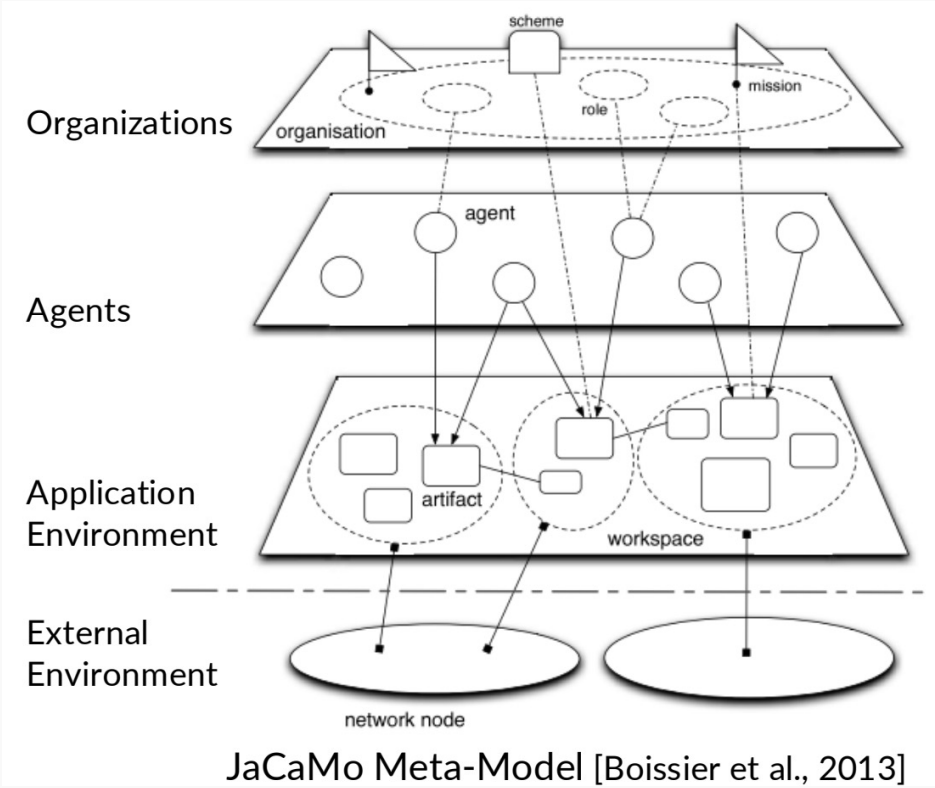
https://bit.ly/webagents-interop

What should be the **role of the Web** in the emerging landscape of Agentic AI?

What are the **relevant standards** within and maybe also outside the W3C?

What are the **standardization gaps** (if any)?

A Layered View of Multi-Agent Systems



JaCaMo Meta-Model [Boissier et al., 2013]

Agents

Goals, reasoning and autonomy

Environment

Resources, state and controls

Interaction

Messages and coordination

Organization

Roles, rules and commitments

A shared lens for goals, architectural patterns and standards

Web Agent Design Goals: Agent Level

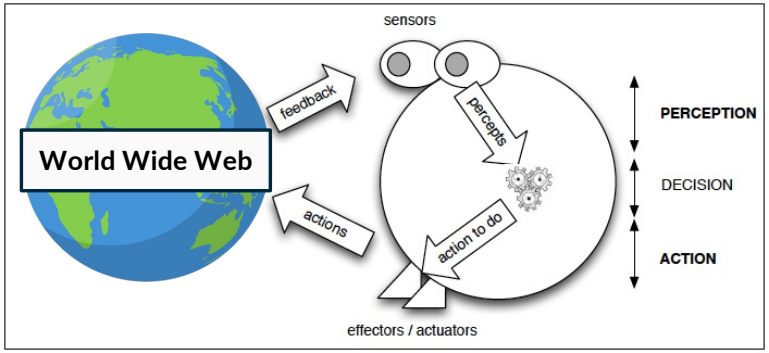
Defining characteristic of agents

Enables agents to discover and interact with one another on the open Web

Fundamental not only to intelligent agents but also to Web user agents in general

Agent-level Design Goals

Design Goal	*Description*
Situatedness	The agent interacts with its hypermedia environment directly through perception and action, and responds in a timely fashion to sensory input.
Embodiment	The agent is represented explicitly in the hypermedia environment, allowing end-users and other agents to discover and interact with it.
Value Alignment	The agent acts in ways that are consistent with the goals, preferences, and interests of its end-user, and that respect human values, ethical principles, societal norms, and fundamental human rights.



[Bordini *et al.*, 2007]

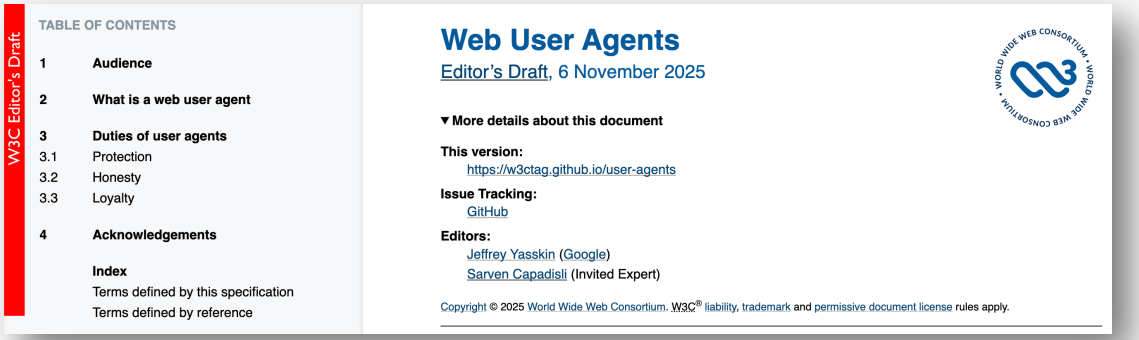

W3C Editor's Draft

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Terms defined by this specification
Terms defined by reference

Web User Agents

Editor's Draft, 6 November 2025

▼ More details about this document

This version:
https://w3ctag.github.io/user-agents

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GitHub

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https://w3ctag.github.io/user-agents/

Rafel Bordini et al., *Programming Multi-Agent Systems in AgentSpeak using Jason*. John Wiley & Sons, 2007.

Web Agent Design Goals: System Level

Central design goals for deploying MAS at scale on the open Web

Observation: core strengths of the Web as a world-wide, open, and long-lived system

Hypothesis: MAS can inherit these strengths if they align with the Web architecture

Necessary for agents to perceive their hypermedia environments

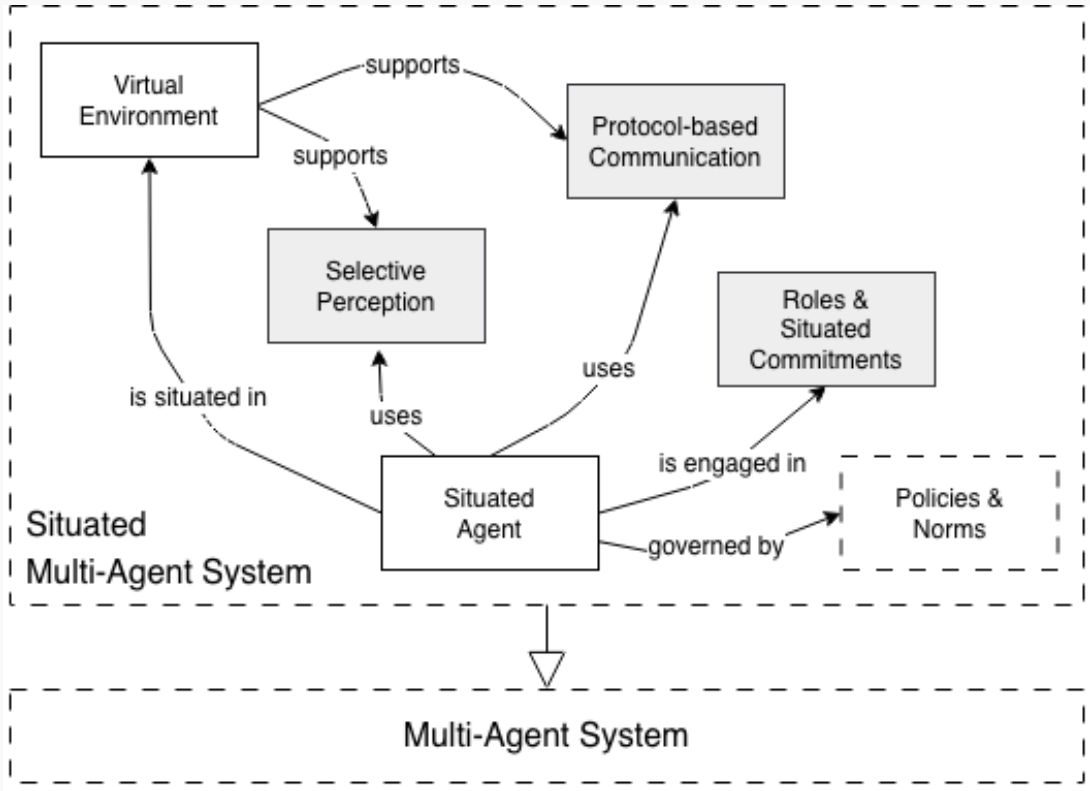
Prerequisite for accountability, explainability, and trust

Critical for agents on the open Web

System-level Design Goals

Design Goal	*Description*
Scalability	The system can support growing numbers of end-users, agents, tools, and other resources across geographical and organizational boundaries.
Interoperability	The system uses Web standards to enable the integration of components developed independently, and to support communication and interaction with other systems.
Extensibility	The system can be expanded with new functionality and resources.
Evolvability	The system can accommodate changes at run time without disrupting existing functionality.
Discoverability	The system enables end-users and agents to discover the rest of the system starting from a single entry URL.
Resource Monitoring	The system enables the selective monitoring of resources, allowing agents and end-users to perceive and react to relevant changes.
Transparency	The system enables the representation, inspection, and reproduction of autonomous behaviors and interactions.
Security	The system provides sufficient assurance for the autonomous discovery of and interaction with agents, tools, and other resources.

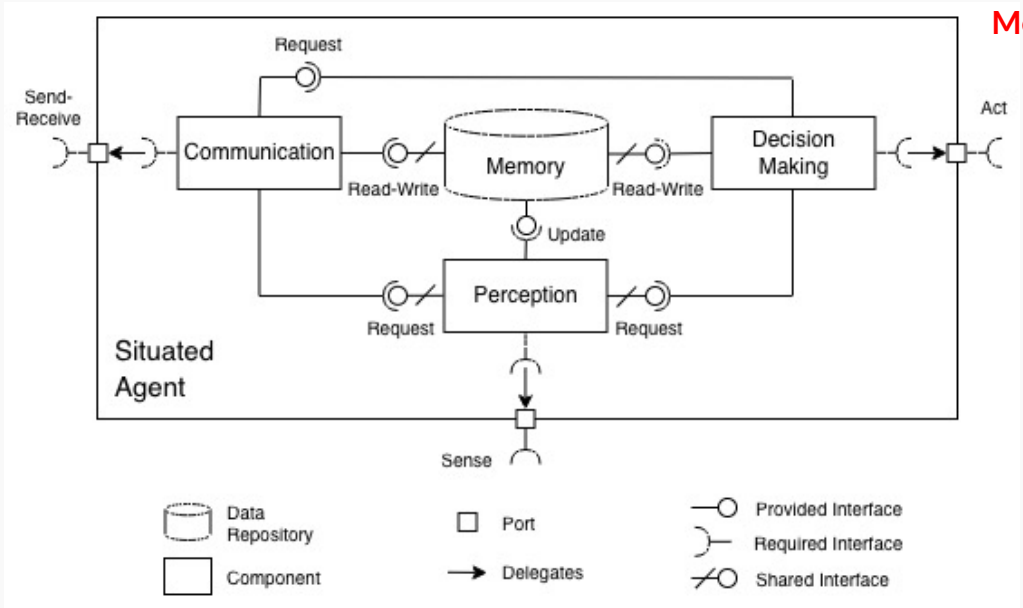
Architectural Patterns for MAS



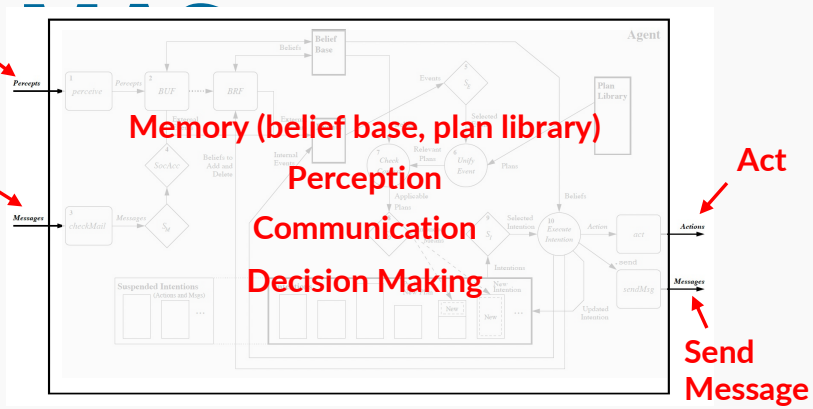


Danny Weyns: *Architecture-based Design of Multi-Agent Systems*. Springer. 2010.

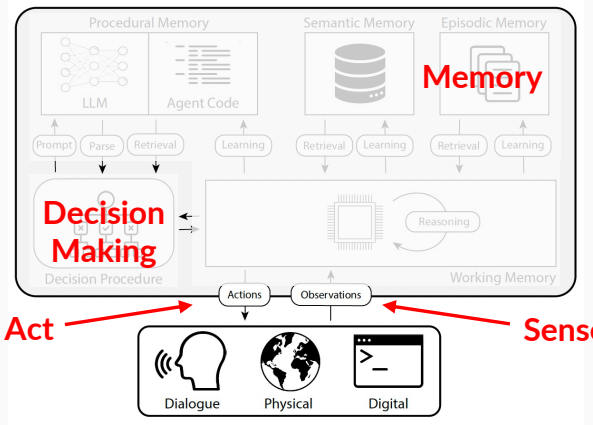
Architectural Patterns for



Note: not all components or their interfaces may be needed in a given implementation



Jason Architecture for BDI Agents
[Bordini *et al.*, 2007]

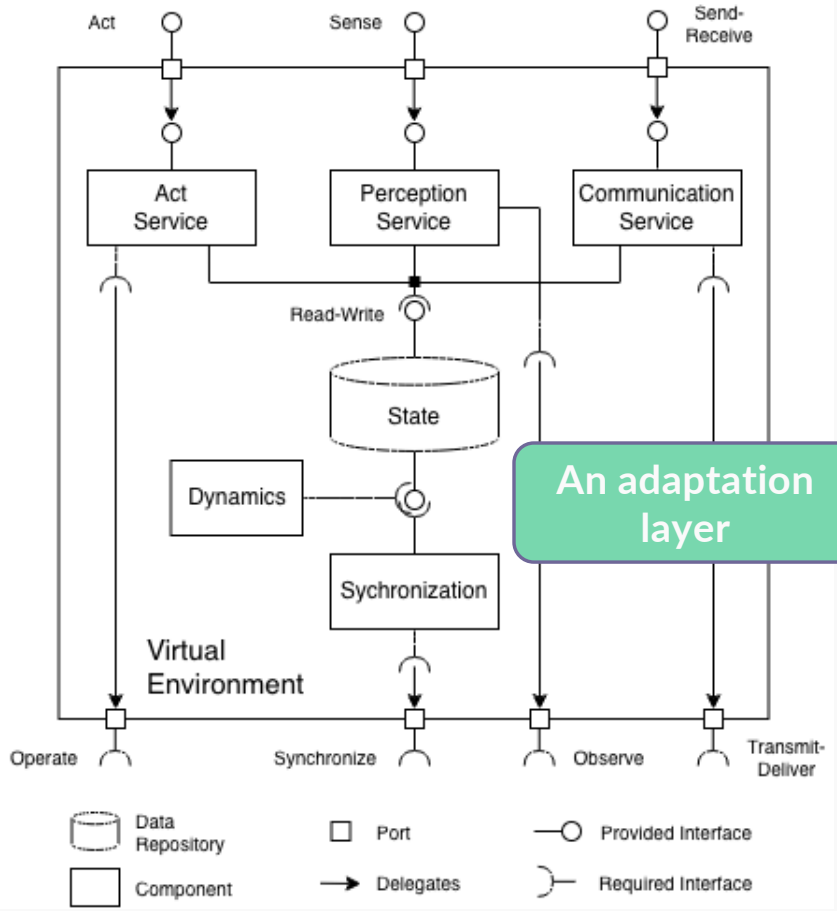


Cognitive Architectures for Language Agents (CoALA)
[Summers *et al.*, 2024]

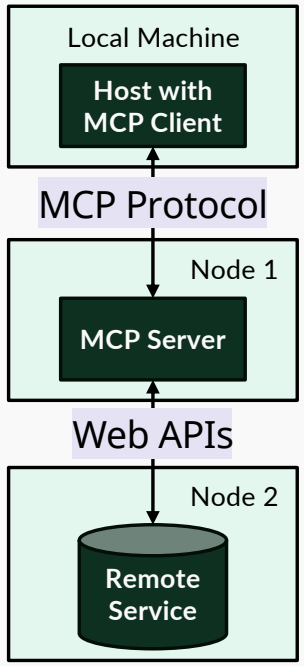
Rafel Bordini et al., *Programming Multi-Agent Systems in AgentSpeak using Jason*. John Wiley & Sons, 2007.
Theodore R. Sumers et al., *Cognitive Architectures for Language Agents*, TMLR, 2024. https://arxiv.org/abs/2309.02427

Architectural Patterns for MAS

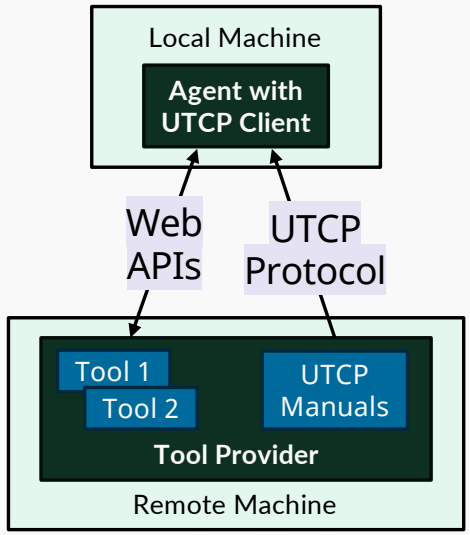
Note: not all components or their interfaces may be needed in a given implementation



An adaptation layer



Model Context Protocol (MCP)



Universal Tool Calling Protocol (UTCP)

MCP example adapted from:
https://modelcontextprotocol.io/specification/2025-06-18/architecture#core-components

Open vs Controlled Environments

Controlled environments

A single curator selects what agents, resources and services exist.

Process meanings and changes are reconciled by the curator.

Access rights and responsibilities are pre-assigned by the curator.

Open environments

The domain of multiple curators.

Processes reflect different perspectives and evolve independently.

Rights and responsibilities established at runtime through diverse means.

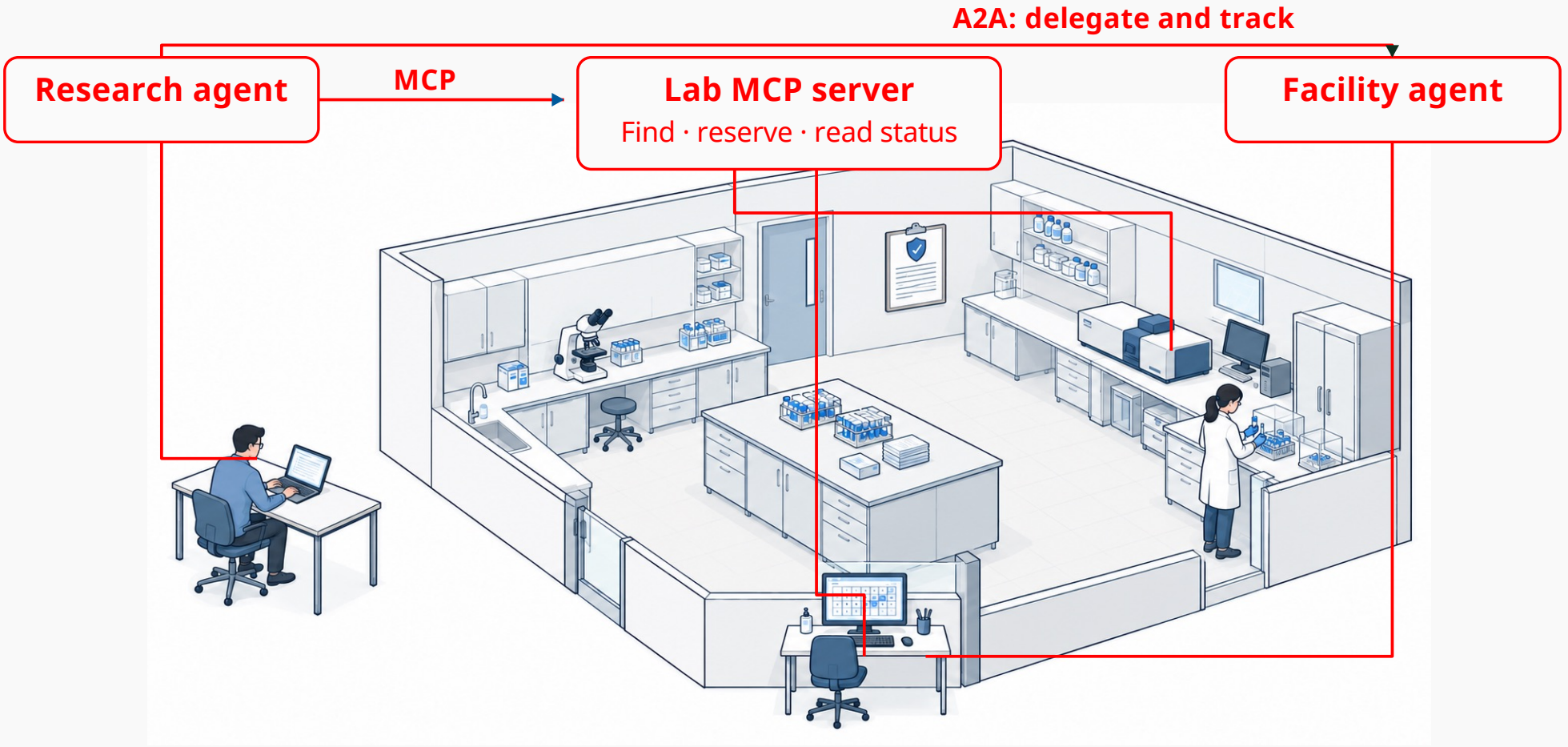
Openness demands agents that can discover and work with unfamiliar resources, services and other agents.

A laboratory task

“Find a suitable instrument, reserve it within my budget, and tell me when it is ready.”



The MCP/A2A laboratory



How MCP and A2A support the laboratory

Laboratory concern	MCP contribution	A2A contribution
Find suitable equipment	Tools and resources expose catalogues, capabilities and availability.	A facility agent finds and proposes suitable equipment.
Understand processes and policies	Resources, tool descriptions and policy queries provide relevant information.	A facility agent explains or manages procedures and checks requirements.
Respond to change	Notifications or queries expose changes; application logic assesses their impact.	Task updates communicate effects on delegated work; agents coordinate a response.

How Open is this?

MCP and A2A offer	Openness requires
Descriptions of known tools and agents	Finding and assessing capabilities across unfamiliar environments
Tool use and task delegation	Understanding how to act within local processes, policies and commitments
Environmental context and task updates	Relating changes to ongoing work and deciding how to adapt

MCP and A2A support interaction.
Effective action depends on context about the environment.

Describing interaction with WoT

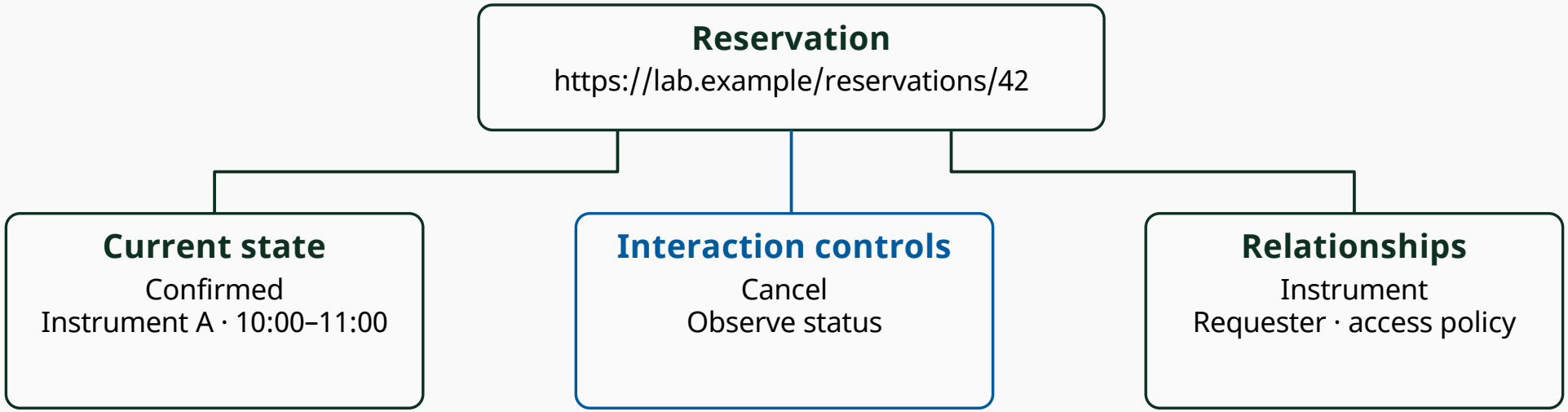
Descriptions connect capabilities to concrete interactions



	The lamp	The instrument
Property	On / off	Availability
Action	Toggle	Start analysis
Event	Lamp failure	Analysis completed

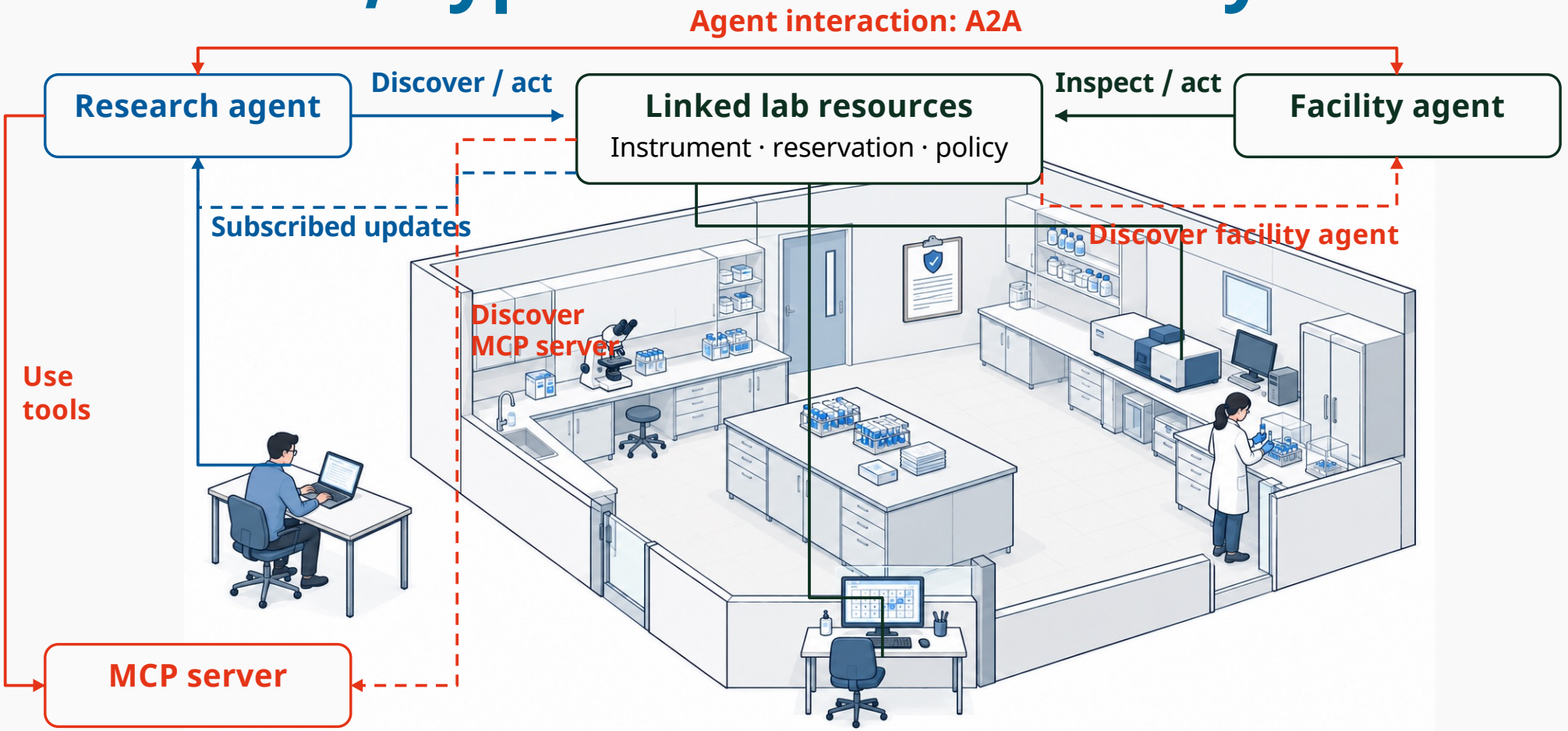
WoT Discovery finds descriptions. Forms describe access to properties, actions and events.

Resources combine state and controls



A resource stereotype supplies shared expectations.
Its representation supplies current state and concrete controls.

The Web/hypermedia laboratory

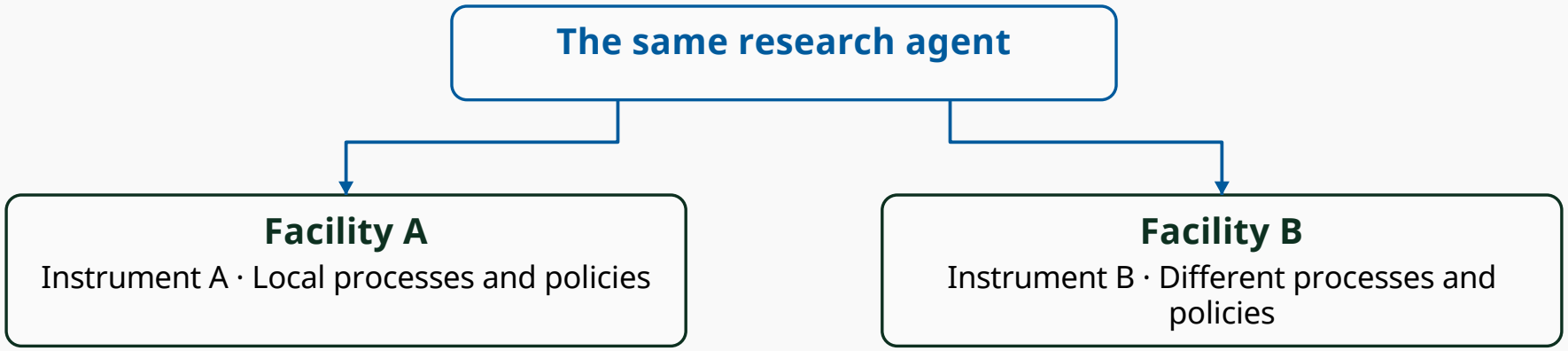


How the Web supports the laboratory

Laboratory concern	Web / hypermedia contribution
Find suitable equipment	Follow linked descriptions of instruments, capabilities and availability, and discover relevant interfaces.
Understand processes and policies	Inspect linked requirements and policies; use controls offered for the resource’s current state.
Respond to change	Observe resource updates, inspect their effect on the reservation and identify available next steps.

Resource navigation, tool use and agent delegation can refer to the same shared environment.

Working with an unfamiliar laboratory



MCP / A2A

Inspect tool and resource descriptions, or ask the facility agent about its procedures.

Web / hypermedia

Inspect linked resources and controls describing current state and available interactions.

MCP/A2A can support a shared environment; hypermedia makes it the organizing principle.

Both require interpretation, suitability assessment and authorization.

When ongoing work changes

Both approaches must support awareness, interpretation and response.

Change	What must the agent establish?
Booking policy changes	What changed, does it affect this reservation, and what can I do now?
The instrument fails	What happens to the booking and commitments, and who must act?
Another agent takes over	What is the current state, what was agreed, and what authority does the replacement need?

How do MCP/A2A and Web/hypermedia help agents respond to change?

When ongoing work changes

Both approaches must support awareness, interpretation and response.

Change	For MCP/A2A
Booking policy changes	Read changed policy and booking via resources/tools; infer next steps from descriptions or facility dialogue.
The instrument fails	Query booking and obligations; use recovery tools or delegate resolution to the facility agent.
Another agent takes over	Load persisted task context and booking state; establish authority before continuing through tools or delegation.

MCP resources and notifications can expose change.
The implementation connects observations, state and available actions.

When ongoing work changes

Both approaches must support awareness, interpretation and response.

Change	For Web/Hypermedia
Booking policy changes	Inspect linked policy and reservation; assess the impact and use the controls currently offered.
The instrument fails	Inspect booking state and commitments; follow offered recovery controls or contact the responsible agent.
Another agent takes over	Follow the reservation URI to shared state, commitments and controls; establish the replacement's authority.

Representations explicitly connect state, relationships and controls.
Observation, interpretation and authority still require design.

How the Web support the design goals

Architectural feature	Goals it can support
Linked, recognizable resource descriptions	Discoverability and interoperability
Observable state and explicit interaction controls	Resource monitoring and situatedness
Persistent shared resources and explicit commitments	Transparency and evolvability
Described access requirements and enforced permissions	Security and safeguards for value alignment

These features support the goals; they do not guarantee them.

Why standards matter

LLMs can already access the Web. What do standards add?

From Inference

Interpret what “Book” commits each party to

Enact local processes from descriptions

Reconstruct ongoing work from conversations

To Shared Conventions

Use shared definitions of reservation states and commitments

Represent requirements and interaction controls explicitly

Maintain shared references to current state and commitments

Shared conventions reduce inference and bespoke integration across independently developed agent systems.

Building on an existing standards ecosystem

Role	Examples	Contribution
Already used by MCP/A2A	OAuth; OpenID Connect; OpenAPI security schemes in A2A	Authorization, identity-provider discovery and security requirements
Additional environment descriptions	RDF/RDFS/OWL, JSON-LD, SHACL; WoT TD and Discovery	Shared models, validation, discovery and interaction descriptions
Policies, evidence and updates	ODRL, Verifiable Credentials, PROV, WebSub	Requirements and duties, verifiable claims, provenance and updates
Further non-W3C specifications	OpenAPI / Arazzo AsyncAPI / CloudEvents XACML	API operations and workflows Event channels and envelopes Access-policy evaluation

Compose these foundations into coherent environments.
Adopting a protocol does not establish the whole architecture.

The Web as a platform for agents

The Web helps us move from connecting agents to capabilities towards designing the shared environments in which they act.

Web architecture provides foundations for doing this across independent curators.